Analytical Cume
2 October 2015

1 (30 points)
1a. If 15 mL of 0.060M $AgNO_3$ solution in the beaker is titrated with 0.030 M NaBr, Find the equivalence point volume.

b. If a silver wire was placed in the titration beaker during the titration, such that a reference electrode is connected to the black lead of the meter and the red lead is connected to a silver wire or:

S.C.E. || titration solution| Ag(s)

Calculate the voltmeter reading at 15 mL of 0.030 M sodium bromide is added.

2 (40 points)
2a. Theoretical Calculations and estimations:
Using the two half reactions below, calculate the standard E^o_{rxn} (in Volts) between Cu^{2+} and $Zn_{(s)}$

$Cu^{2+} + 2e^- \leftrightarrow Cu_{(s)}$ $E^o = +0.339$ V
$Zn^{2+} + 2e^- \leftrightarrow Zn_{(s)}$ $E^o = -0.762$ V

2b. Write the spontaneous reaction that would occur between zinc metal and copper sulfate.

A piece of zinc metal is dropped into the copper sulfate solution, the reaction occurs, and solid metal zinc is left after the reaction has stopped.

2c. Write the equilibrium expression (K=) for the chemical reaction written in 2b.

2d. Find the value of the equilibrium constant K for the reaction in 2b .

2e. If the starting concentration of copper sulfate is 0.05 M.
Write the charge balance for the solution after the reaction has stopped.

2f. Write a mass balance for sulfate.

2g. Find the equilibrium concentration for Cu^{2+} ion.

3 (30 points)

3. Using cell potentials to calculate solution concentrations.

Consider the following cell

(black) Pt | H_2(g, 1.00 bar) | H^+ (aq, pH 3.6) | | Cl^-(?.?M), AgCl(s) | Ag (red)

If E_{meter} = +0.511 V what is the concentration of the chloride ion in the right-hand beaker?

Half-Reaction

	E^o
$Ag^+ + e^- \leftrightarrow Ag(s)$	+0.799 V
$Fe^{3+} + 1e^- \leftrightarrow Fe^{2+}$	+0.771
$I_2 + 2e^- \leftrightarrow 2I^-$	+0.620
$Cu^{2+} + 2e^- \leftrightarrow Cu(s)$	+0.339
$Zn^{2+} + 2e^- \leftrightarrow Zn(s)$	−0.762
$AgCl(s) + 1e^- \leftrightarrow Ag(s) + Cl^-$	+0.222
$UO_2^{2+} + 4H^+ + 2e^- \leftrightarrow U^{4+} + 2H_2O$	+0.273
$2H^+ + 2e^- \leftrightarrow H_2$ (g)	+0.0
$Hg_2Cl_2 + 2e^- \leftrightarrow 2Hg(l) + 2Cl^-$	0.241 (calomel)

Formulas

$E = E_{red} - E_{black}$

For the reaction:

$\gamma O + ne- \leftrightarrow \beta R$

The thermodynamic description is:

$$E = E^o - \frac{0.0591}{n}\log\frac{[R]^\beta}{[O]^\gamma}$$

$$E^o = \frac{0.0591}{n} logK$$

F = 96,483 coul/mol e^-